

Experiment – 6

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(A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

Solution:

```
CREATE TABLE Sales
```

```
(
    sale_date DATE,
    fruit VARCHAR(20),
    sold_num INT,
    PRIMARY KEY (sale_date, fruit)
);
```

```
INSERT INTO Sales
```

```
VALUES
```

```
('2020-05-01', 'apples', 10),
```

```
( '2020-05-01', 'oranges', 8),

( '2020-05-02', 'apples', 15),
( '2020-05-02', 'oranges', 15),

( '2020-05-03', 'apples', 20),
( '2020-05-03', 'oranges', 0),

( '2020-05-04', 'apples', 15),
( '2020-05-04', 'oranges', 16);
```

```
SELECT sale_date,
       SUM(
         CASE
           WHEN fruit = 'apples' THEN sold_num
           ELSE -sold_num
         END
       ) AS diff
FROM Sales
GROUP BY sale_date
ORDER BY sale_date;
```

(B)

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- **id** represents the current node.
- **p_id** represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Tree

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Expected Output

id	Type
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf

Solution:

```
CREATE TABLE Tree
```

```
(
    id INT PRIMARY KEY,
    p_id INT
);
```

```
INSERT INTO Tree
```

```
VALUES
```

```
(1, NULL),
```

```
(2, 1),
```

```
(3, 1),
```

```
(4, 2),
```

```
(5, 2);
```

```
SELECT ID,
```

```
    CASE
```

```
        WHEN P_ID IS NULL THEN 'ROOT'
```

```
        WHEN ID NOT IN (
```

```
            SELECT P_ID
```

```
            FROM TREE
```

```
            WHERE P_ID IS NOT NULL
```

```
        ) THEN 'LEAF'
```

```
        ELSE 'INNER'
```

```
    END AS "TYPE"
```

```
FROM TREE;
```